web2_bordered (1).pdf
PDF

Gpt aa file mathi badha practical code study karine mane details ma badha practical code kema run karva kema lakhva starting to end sudhi explain kar ane aa pdf web-II programming code che je visual studio code, node. Js Jason na che to te kemathi run thase and kya write karvana ae mane sikhav

Bhai mast 👌 tame je PDF mukyu che e Web Designing-II (Node.js + JSON + AJAX) na practicals che. Hu tane starting thi end sudhi step-by-step full samjavi dau — ke:

👉 Code kya lakhvu
👉 VS Code ma kem run karvu
👉 Node.js kem use karvu
👉 HTML/JSON kem open karvu

🔥 STEP 1: Setup (IMPORTANT)

1. Install Node.js

- Google → "Node.js download"
- Install kari de

Check:

```
node -v
```

---

2. Install Visual Studio Code

- Install kari de

---

3. Folder banav

Example:

```
Web2_Practical/
```

🧠 BASIC RULE (Yaad rakh)

Type	File	Run kem karvu
Node.js	`.js`	terminal ma `node filename.js`
HTML	`.html`	browser ma open
JSON	`.json`	data file (direct run nathi thatu)

🟢 PRACTICAL 15 – Simple Node Program

File:

`app.js`

```
console.log("Welcome to Node.js");
console.log("Node Version:", process.version);
```

Run:

```
node app.js
```

👉 Output terminal ma avse

🟢 PRACTICAL 16–17 (SERVER)

File:

`server.js`

```
const http = require('http');

http.createServer(function(req,res){
    res.write("Hello Node.js Server");
    res.end();
}).listen(3000);

console.log("Server running at http://localhost:3000");
```

Run:

```
node server.js
```

👉 Browser open:

```
http://localhost:3000
```

🟢 PRACTICAL 18 – Module

File 1:

`pr4.js`

```
exports.add = function(a,b){
    return a + b;
}
```

File 2:

`main.js`

```
const math = require('./pr4');
console.log("Addition:", math.add(10,5));
```

Run:

```
node main.js
```

🟢 PRACTICAL 19 – Query String

👉 Browser ma:

```
http://localhost:3000/?name=raj&city=Ahmedabad
```

👉 Code same PDF jevu

🟢 PRACTICAL 20 – Routing

👉 URL change kari check kar:

- `/` → Home
- `/about` → About

🟢 PRACTICAL 21 – File System

File: `fs.js`

```
const fs = require('fs');

fs.writeFile('demo.txt','Hello',(err)=>{
    console.log("File created");

    fs.readFile('demo.txt','utf8',(err,data)=>{
        console.log(data);
```

```
        fs.appendFile('demo.txt',' New Data',(err)=>{
            console.log("Appended");

            fs.rename('demo.txt','new.txt',(err)=>{
                console.log("Renamed");

                fs.unlink('new.txt',(err)=>{
                    console.log("Deleted");
                });

            });

        });

    });

});
```

Run:

```
node fs.js
```

🟢 PRACTICAL 22 – HTML File Server

File 1:

`index.html`

File 2:

`server.js`

👉 Run:

```
node server.js
```

👉 Browser:

```
localhost:3000
```

🟢 PRACTICAL 23 – OS Info

👉 Same server code but:

```
const os = require('os');
```

🟢 PRACTICAL 24 – FULL SERVER PROJECT

👉 Aa thodu important che exam mate 🔥

Features:

- Home
- About
- Create file
- Read file
- Delete file
- User data

👉 Run:

```
node server.js
```

👉 Test URLs:

```
/about
/create
/read
/delete
/user?name=raj&city=ahmedabad
```

🟡 PRACTICAL 25–30 (JSON + HTML)

👉 Aa badha simple che

Rule:

- `.html` file banav
- Browser ma open kar

Example:

```
<script>
let emp = {"name":"Amit"};
document.write(emp.name);
</script>
```

👉 Run → Double click HTML file

🟡 PRACTICAL 31 – Fetch JSON

Files:

- `index.html`
- `file2.json`

👉 IMPORTANT:
❌ Direct open na thay (CORS issue)

👉 Use:
VS Code Extension → Live Server

🔵 PRACTICAL 32 – AJAX PROJECT

Files:

- index.html
- script.js
- students.json
- style.css

👉 Run:
✔ Right click → "Open with Live Server"

👉 Browser ma:

- ID enter kar
- Button click → data show

⚡ SHORTCUT SUMMARY

Work	Tool
Node.js code	Terminal
HTML	Browser
JSON + AJAX	Live Server

💡 IMPORTANT TIPS (Exam mate)

✔ Always `require()` use
✔ Server ma `res.end()` jaruri
✔ `listen(3000)` must
✔ JSON = key:value format
✔ AJAX = XMLHttpRequest

🚀 FINAL FLOW (Yaad rakh)

1. VS Code open
2. Folder open
3. File banav (.js / .html)
4. Code paste
5. Terminal open
6. Run:

```
node filename.js
```

7. Browser open (localhost)

👍 Bhai jo tu kahe to hu:

- Badha practical viva Q&A pan api dau
- Athva ek-ek practical ne simple Gujarati/Kathiyawadi ma explain kari dau

Bol bas 😎

First sikhav ke practical 15 no code kya kahvo ane run kya karvo

Saru bhai 👌 chalo ekdam basic thi start kariye — Practical 15 step-by-step (ekdum beginner level ma samjavu chu).

🟢 PRACTICAL 15 – Code kya lakhvu ane run kya karvu

🧠 Aa practical ma su che?

👉 Simple Node.js program che je message print kare che

🪜 STEP 1: Visual Studio Code open karo

1. VS Code open karo
2. "File" → "Open Folder"
3. New folder banavo
   👉 Name: `web2`

🪜 STEP 2: File banavo

1. Left side explorer ma right click karo
2. Click → New File
3. Name lakho:

```
app.js
```

👉 `.js` bahu important che (Node.js file mate)

🪜 STEP 3: Code lakho

Aa code exact lakhvo:

```
console.log("Welcome to Node.js");
console.log("Node Version:", process.version);
```

👉 Save karo (Ctrl + S)

🪜 STEP 4: Node.js check karo

VS Code ma terminal open karo:

👉 Shortcut:

```
Ctrl + `
```

Pachi lakh:

```
node -v
```

👉 Version ave to perfect 👍

🪜 STEP 5: Code RUN karo

Terminal ma lakh:

```
node app.js
```

🎯 OUTPUT su avse?

Terminal ma aa dekhase:

```
Welcome to Node.js
Node Version: v18.x.x
```

❗ Common mistake (avoid karje)

❌ `.js` ni jagyae `.txt` file banavi
❌ `node app.js` ni jagyae `app.js` lakhvu
❌ Node install na hoy

🧠 Easy yaad rakh trick

👉 "JS file → Node thi run → Output terminal ma"

🚀 Short summary

1. VS Code open
2. `app.js` file banav
3. Code lakh
4. Terminal open
5. `node app.js` run

👍 Have tu try kar ane bol — pachhi hu Practical 16 (server) pan mast rite samjavi dau (thodu interesting che 😎)

Next practical 16 pan aa same rite sikhav

Perfect bhai 😎 have Practical 16 (Server) sikhiye — aa thodu interesting che, karan ke aa ma tame browser ma output joy shakso 🔥

🟢 PRACTICAL 16 – Simple HTTP Server

🧠 Aa practical ma su che?

👉 Node.js thi ek server banavso
👉 Browser ma message show karso

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`) open rakho

🪜 STEP 2: New file banavo

File name:

```
server.js
```

🪜 STEP 3: Code lakho

Aa code exact lakhvo:

```
const http = require('http');

http.createServer(function(req,res){
    res.write("Hello Node.js Server");
    res.end();
```

```
}).listen(3000);

console.log("Server running at http://localhost:3000");
```

👉 Save karo

🪜 STEP 4: Terminal open karo

Shortcut:

```
Ctrl + `
```

🪜 STEP 5: Run karo (Node.js thi)

```
node server.js
```

👉 Terminal ma avse:

```
Server running at http://localhost:3000
```

🪜 STEP 6: Browser ma check karo 🌐

👉 Chrome / Edge open karo
👉 Address bar ma lakh:

```
http://localhost:3000
```

🎯 OUTPUT su avse?

👉 Screen par avse:

```
Hello Node.js Server
```

🧠 Code samjuti (simple language ma)

1.

```
const http = require('http');
```

👉 HTTP module load kare che (server banava mate)

2.

```
http.createServer(function(req,res){
```

👉 Server banave che

👉 `req` = request

👉 `res` = response

3.

```
res.write("Hello Node.js Server");
```

👉 Browser ma message mokle che

4.

```
res.end();
```

👉 Response finish kare che (IMPORTANT ❗)

5.

```
.listen(3000);
```

👉 Server port 3000 par chale che

❗ Important note

👉 Server chalti vakhat terminal bandh na karvu
👉 Stop karva mate:

```
Ctrl + C
```

🚀 Short summary

1. `server.js` file banav
2. Code lakh
3. `node server.js` run
4. Browser → `localhost:3000`

👍 Bhai aa samji gayu hoy to mast 🔥
Have bol — hu Practical 17 (HTML server) pan same style ma sikhavu? 😄

Yes practical 17

Chalo bhai 😎 have next level — Practical 17 (HTML Server)
Aa ma simple text ni jagyae proper HTML page browser ma show thase 🔥

🟢 PRACTICAL 17 – HTTP Server with HTML Response

🧠 Aa practical ma su che?

👉 Node.js server banavso
👉 HTML code browser ma display karso

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`) use karvo

🪜 STEP 2: File banavo

File name:

```
server2.js
```

🪜 STEP 3: Code lakho

Aa code exact lakhvo:

```
const http = require('http');

http.createServer(function(req,res){
    res.writeHead(200, {'Content-Type': 'text/html'});
    res.write("<h1>Welcome to Node.js Web Server</h1>");
    res.end();
}).listen(3000);

console.log("Server running at http://localhost:3000");
```

👉 Save karo

🪜 STEP 4: Terminal open karo

```
Ctrl + `
```

🪜 STEP 5: Run karo (Node.js thi)

```
node server2.js
```

🪜 STEP 6: Browser ma open karo 🌐

```
http://localhost:3000
```

🎯 OUTPUT su avse?

👉 Screen par heading avse:

👉 Welcome to Node.js Web Server

(Heading mota size ma hase — karan ke `<h1>` use karyu che)

🧠 Code samjuti (easy language ma)

1.

```
res.writeHead(200, {'Content-Type': 'text/html'});
```

👉 Browser ne kahe che:
➡️ “Aa HTML content che”

2.

```
res.write("<h1>Welcome...</h1>");
```

👉 HTML tag use kari ne design show kare che

3.

```
res.end();
```

👉 Response complete kare che (very important ❗)

❗ Practical 16 vs 17 difference

Practical 16	Practical 17
Plain text	HTML output
Simple message	Design (h1, p, etc.)
No content-type	`text/html`

⚠️ Common mistake

❌ `writeHead` bhuli jav
❌ Port same hoy ane already run hoy
👉 Error ave → pehla server stop:

```
Ctrl + C
```

🚀 Short summary

1. `server2.js` file banav

2. HTML code `res.write()` ma lakh
3. Run → `node server2.js`
4. Browser → `localhost:3000`

🔥 Bhai aa sudhi tu Node.js basic pakdi lidhu che!
Have next ma Practical 18 (Module) aavse — thodu programming logic strong karse 💪

Bol sikhu? 😄

Practical 18

Chalo bhai 😎 have Practical 18 (User Defined Module) — aa thodu powerful concept che 💪 (exam ma pan ave che 🔥)

🟢 PRACTICAL 18 – Create User Defined Module

🧠 Aa practical ma su che?

👉 Tame ek file ma function banavso
👉 Biji file ma ene use karso

➡️ Simple words ma:
"Ek file mathi code export karo ane bija file ma use karo"

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`) ma kaam karisu

🪜 STEP 2: 2 file banavani che

📄 File 1:

```
pr4.js
```

📄 File 2:

```
main.js
```

🪜 STEP 3: pr4.js ma code lakh

```
exports.add = function(a,b){
    return a + b;
}
```

👉 Save karo

🪜 STEP 4: main.js ma code lakh

```
const math = require('./pr4');

console.log("Addition:", math.add(10,5));
```

👉 Save karo

🪜 STEP 5: Terminal open karo

```
Ctrl + `
```

🪜 STEP 6: Run karo (Node.js thi)

```
node main.js
```

🎯 OUTPUT su avse?

```
Addition: 15
```

🧠 Code samjuti (easy way ma)

🔹 pr4.js

```
exports.add = function(a,b)
```

👉 `exports` = function bahar mokalva mate

👉 `add` = function name

🔹 main.js

```
const math = require('./pr4');
```

👉 `require()` thi file import kari

⚠️ `.js` lakhvani jarur nathi

```
math.add(10,5)
```

👉 pr4.js ni function call kari

❗ Important rules

✔ File name correct hovu joiye (`./pr4`)
✔ Same folder ma banavi
✔ `exports` lakhvu jaruri

🧠 Real life example

👉 pr4.js = calculator 📟
👉 main.js = user 👤

User → calculator ne kahe → addition kar

🚀 Short summary

1. 2 file banav (pr4.js, main.js)
2. Function export karo
3. `require()` thi import karo
4. Run → `node main.js`

🔥 Bhai aa concept samji gayo to Node.js easy thai jashe!

Have bol — hu next Practical 19 (Query String URL) pan aa same style ma sikhavu? 😄

Practical 19

Chalo bhai 😎 have Practical 19 (Query String) — aa thodu mast concept che 🔥 (URL mathi data levanu)

🟢 PRACTICAL 19 – Read Query String from URL

🧠 Aa practical ma su che?

👉 Browser na URL mathi data levu
👉 Example:

```
http://localhost:3000/?name=raj&city=Ahmedabad
```

➡️ Aa ma:

- `name = raj`
- `city = Ahmedabad`

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: File banavo

```
query.js
```

🪜 STEP 3: Code lakho

```
const http = require('http');
const url = require('url');

http.createServer(function(req,res){
    const q = url.parse(req.url, true).query;

    res.write("Name: " + q.name + "\n");
    res.write("City: " + q.city);

    res.end();
}).listen(3000);
```

```
console.log("Server running at http://localhost:3000");
```

👉 Save karo

🪜 STEP 4: Terminal open karo

```
Ctrl + `
```

🪜 STEP 5: Run karo (Node.js thi)

```
node query.js
```

🪜 STEP 6: Browser ma open karo 🌐

👉 Aa exact URL lakhjo:

```
http://localhost:3000/?name=raj&city=Ahmedabad
```

🎯 OUTPUT su avse?

Screen par:

```
Name: raj
City: Ahmedabad
```

🧠 Code samjuti (easy way ma)

🔹 1.

```
const url = require('url');
```

👉 URL mathi data levava mate module

🔹 2.

```
url.parse(req.url, true).query;
```

👉 URL ne parse kari ne data extract kare che

🔹 3.

```
q.name
q.city
```

👉 URL ma je value api e access kare che

❗ Important samjan

👉 URL format:

```
?key=value&key=value
```

Example:

```
?name=raj&city=Ahmedabad
```

⚠️ Common mistake

❌ `?` bhuli jav
❌ `&` na mukvu
❌ server run na hoy

🧠 Real life example

👉 URL = form data
👉 Server = data read kare

🚀 Short summary

1. `query.js` file banav
2. Code lakh
3. `node query.js` run
4. Browser ma URL ma data pass kar

🔥 Bhai aa bahu important che (future ma API ma use thase)

Have bol — next Practical 20 (Routing) sikhavu? 😎

Practical 20

Chalo bhai 😎 have Practical 20 (Routing) — aa Node.js ma ekdam important concept che 🔥

🟢 PRACTICAL 20 – Routing Example

🧠 Aa practical ma su che?

👉 URL change karo ane different output male

➡️ Example:

- `/` → Home Page
- `/about` → About Page
- biju kai pan → Page Not Found

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: File banavo

```
route.js
```

🪜 STEP 3: Code lakho

```
const http = require('http');

http.createServer(function(req,res){

    if(req.url == "/"){
        res.write("Home Page");
    }
    else if(req.url == "/about"){
        res.write("About Page");
    }
    else{
        res.write("Page Not Found");
    }

    res.end();

}).listen(3000);

console.log("Server running at http://localhost:3000");
```

👉 Save karo

🪜 STEP 4: Terminal open karo

```
Ctrl + `
```

🪜 STEP 5: Run karo (Node.js thi)

```
node route.js
```

🪜 STEP 6: Browser ma check karo 🌐

👉 Case 1:

```
http://localhost:3000/
```

➡️ Output: Home Page

👉 Case 2:

```
http://localhost:3000/about
```

➡️ Output: About Page

👉 Case 3:

```
http://localhost:3000/xyz
```

➡️ Output: Page Not Found

🧠 Code samjuti (easy language ma)

🔹 1.

```
req.url
```

👉 Browser ma je URL lakhyu e check kare che

🔹 2.

```
if(req.url == "/")
```

👉 Home page mate condition

🔹 3.

```
else if(req.url == "/about")
```

👉 About page mate

🔹 4.

```
else
```

👉 Bijun badhu → error message

❗ Important note

👉 Server run hoy tyare j browser ma kaam karse
👉 Stop karva:

```
Ctrl + C
```

🧠 Real life example

👉 Routing = Website na alag-alag page

URL	Page
`/`	Home
`/about`	About
`/contact`	Contact

🚀 Short summary

1. `route.js` file banav
2. `req.url` thi check kar
3. Condition muk
4. `node route.js` run
5. Browser ma different URL try kar

🔥 Bhai aa concept samji gayo to tu mini website banaavi shake che!

Have bol — next Practical 21 (File System) sikhavu? 😎

Practical 21

Chalo bhai 😎 have Practical 21 (File System) — aa thodu powerful che 🔥 (file create, read, delete badhu ek j code ma)

🟢 PRACTICAL 21 – File System (fs module)

🧠 Aa practical ma su che?

👉 File sathe kaam karvu:

- Create file
- Read file
- Append (data add)
- Rename file
- Delete file

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: File banavo

```
fs.js
```

🪜 STEP 3: Code lakho

👉 Aa code full lakhjo (same sequence ma)

```
const fs = require('fs');

// 1. Create File
fs.writeFile('demo.txt', 'This is Node.js File System Demo', function(err)
    if (err) throw err;
    console.log("File created successfully");

    // 2. Read File
```

```
    fs.readFile('demo.txt', 'utf8', function(err, data) {
        if (err) throw err;
        console.log("File Content:", data);

        // 3. Append Data
        fs.appendFile('demo.txt', '\nNew data added', function(err) {
            if (err) throw err;
            console.log("Data appended");

            // 4. Rename File
            fs.rename('demo.txt', 'newdemo.txt', function(err) {
                if (err) throw err;
                console.log("File renamed");

                // 5. Delete File
                fs.unlink('newdemo.txt', function(err) {
                    if (err) throw err;
                    console.log("File deleted");
                });

            });

        });

    });

});
```

👉 Save karo

🪜 STEP 4: Terminal open karo

```
Ctrl + `
```

🪜 STEP 5: Run karo (Node.js thi)

```
node fs.js
```

🎯 OUTPUT su avse?

Terminal ma:

```
File created successfully
File Content: This is Node.js File System Demo
Data appended
File renamed
File deleted
```

🧠 Step-by-step samjuti

🔹 1. Create File

```
fs.writeFile(...)
```

👉 `demo.txt` file create kare che

🔹 2. Read File

```
fs.readFile(...)
```

👉 File nu content read kare

🔹 3. Append

```
fs.appendFile(...)
```

👉 File ma navo data add kare

🔹 4. Rename

```
fs.rename(...)
```

👉 File nu naam badle

🔹 5. Delete

```
fs.unlink(...)
```

👉 File delete kare

❗ Important samjan

👉 Aa badha operations sequence ma chalta hoy che
👉 Etle ek bijani andar lakhya che (nested)

⚠️ Common mistake

❌ File naam wrong
❌ `fs` require na karvu
❌ save na karvu

🧠 Real life example

👉 File system = mobile gallery 📁

Action	Real life
create	photo levi
read	photo joi

Action	Real life
append	edit/add
rename	naam badlavvu
delete	delete karvu

🚀 Short summary

1. `fs.js` file banav
2. fs module use kar
3. operations lakh
4. `node fs.js` run

🔥 Bhai aa practical bahu important che (interview ma pan ave)

Have bol — next Practical 22 (HTML file server) sikhavu? 😎

Practical 22

Chalo bhai 😎 have Practical 22 (HTML File Server) — aa ekdam mast che 🔥
👉 Node.js thi actual HTML file browser ma show karisu

🟢 PRACTICAL 22 – Display HTML File using Server

🧠 Aa practical ma su che?

👉 `.html` file banavi
👉 Node.js server thi load karavi

➡️ Direct `res.write()` karta better method 👌

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: 2 file banavani che

📄 File 1:

```
index.html
```

📄 File 2:

```
server3.js
```

🪜 STEP 3: index.html ma code lakh

```html
<html>
<head>
<title>Node.js Page</title>
</head>
<body>
<h1>Welcome to Node.js</h1>
<p>This page is loaded using Node.js server</p>
</body>
</html>
```

👉 Save karo

🪜 STEP 4: server3.js ma code lakh

```javascript
const http = require('http');
const fs = require('fs');
```

```
http.createServer(function(req,res){

    fs.readFile('index.html', (err,data)=>{
        res.writeHead(200, {'Content-Type':'text/html'});
        res.write(data);
        res.end();
    });

}).listen(3000);

console.log("Server running at http://localhost:3000");
```

👉 Save karo

🪜 STEP 5: Terminal open karo

```
Ctrl + `
```

🪜 STEP 6: Run karo (Node.js thi)

```
node server3.js
```

🪜 STEP 7: Browser ma open karo 🌐

```
http://localhost:3000
```

🎯 OUTPUT su avse?

👉 Proper webpage open thase:

- Heading: Welcome to Node.js
- Paragraph: This page is loaded using Node.js server

🧠 Code samjuti (simple language ma)

🔹 1.

```
const fs = require('fs');
```

👉 File read karva mate

🔹 2.

```
fs.readFile('index.html', (err,data)=>{
```

👉 HTML file read kare che

🔹 3.

```
res.write(data);
```

👉 File no content browser ma mokle che

🔹 4.

```
res.writeHead(200, {'Content-Type':'text/html'});
```

👉 Browser ne kahe che ke aa HTML che

❗ Important note

👉 `index.html` ane `server3.js` same folder ma hova joiye

⚠️ Common mistake

❌ File name mismatch (`Index.html` vs `index.html`)
❌ server run na hoy
❌ path wrong

🧠 Practical 17 vs 22 difference

Practical 17	Practical 22
HTML direct code ma	HTML file mathi
`res.write("<h1>")`	`fs.readFile()`
Simple	Real website style

🚀 Short summary

1. `index.html` banav
2. `server3.js` ma fs use kar
3. `node server3.js` run
4. Browser → localhost

🔥 Bhai aa thi tu real website serve karva shikhyo

Have bol — next Practical 23 (System Info) sikhavu ke directly important 24 (big server project) par jaiye? 😎

Practical 23

Chalo bhai 😎 have Practical 23 (System Information) — aa simple pan mast che 🔥
👉 Server thi tamara system ni info browser ma dekhashe

🟢 PRACTICAL 23 – Display System Information

🧠 Aa practical ma su che?

👉 Computer ni info levani:

- Hostname
- Platform
- Memory

👉 Browser ma show karvani

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: File banavo

```
osinfo.js
```

🪜 STEP 3: Code lakho

```
const http = require('http');
const os = require('os');

http.createServer(function(req,res){
```

```
    res.write("Hostname: " + os.hostname() + "\n");
    res.write("Platform: " + os.platform() + "\n");
    res.write("Free Memory: " + os.freemem());

    res.end();

}).listen(3000);

console.log("Server running at http://localhost:3000");
```

👉 Save karo

🪜 STEP 4: Terminal open karo

```
Ctrl + `
```

🪜 STEP 5: Run karo (Node.js thi)

```
node osinfo.js
```

🪜 STEP 6: Browser ma open karo 🌐

```
http://localhost:3000
```

🎯 OUTPUT su avse?

Example:

```
Hostname: DESKTOP-1234
Platform: win32
Free Memory: 123456789
```

👉 (Tamara system pramane change thase)

🧠 Code samjuti (easy language ma)

🔹 1.

```
const os = require('os');
```

👉 System info mate module

🔹 2.

```
os.hostname()
```

👉 Computer nu naam

🔹 3.

```
os.platform()
```

👉 OS type (windows / linux / etc.)

🔹 4.

```
os.freemem()
```

👉 Free RAM ketli che

❗ Important note

👉 Aa badhu dynamic data che
👉 Every system ma different hase

⚠️ Common mistake

❌ `os` require bhuli jav
❌ server run na hoy
❌ port busy hoy

🧠 Real life example

👉 Aa server = system monitor 🖥️
👉 Je tamne system info batave

🚀 Short summary

1. `osinfo.js` file banav
2. `os` module use kar
3. `node osinfo.js` run
4. Browser ma result joy

🔥 Bhai aa sudhi tu Node.js na almost badha basics cover kari lidha!

Have next ma aave che Practical 24 (FULL SERVER PROJECT) — aa ekdam important + exam favourite 💥

Bol sikhu? 😎

Chalo bhai 😎 have Practical 24 (Full Server Program) — aa sabthi important che 💥
👉 Aa ek mini project che (exam ma pakka ave 🔥)

🟢 PRACTICAL 24 – Complete Server Program

🧠 Aa practical ma su che?

Ek j server ma badhu:

- Home page
- About page
- User data (URL thi)
- File create
- File read
- File delete

👉 Ekdum combo project 💪

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: File banavo

```
server4.js
```

🪜 STEP 3: Full code lakho

👉 Aa code carefully copy karo:

```
const http = require('http');
const url = require('url');
const fs = require('fs');
```

```
http.createServer(function(req,res){

    const q = url.parse(req.url,true).query;

    res.writeHead(200, {'Content-Type':'text/html'});

    if(req.url == "/"){
        res.write("<h1>Welcome to Node.js Server</h1>");
        res.write("<p>Use /about , /create , /read , /delete</p>");
    }

    else if(req.url == "/about"){
        res.write("<h2>This is About Page</h2>");
    }

    else if(req.url.startsWith("/user")){
        res.write("<h2>User Information</h2>");
        res.write("Name: " + q.name + "<br>");
        res.write("City: " + q.city);
    }

    else if(req.url == "/create"){
        fs.writeFile("demo.txt","Node.js File Created",(err)=>{
            if(err) throw err;
        });
        res.write("File Created");
    }

    else if(req.url == "/read"){
        fs.readFile("demo.txt","utf8",(err,data)=>{
            res.write("File Content: " + data);
            res.end();
        });
        return;
    }

    else if(req.url == "/delete"){
        fs.unlink("demo.txt",(err)=>{
            if(err) throw err;
        });
        res.write("File Deleted");
    }

    else{
        res.write("Page Not Found");
    }

    res.end();
```

```
}).listen(3000);

console.log("Server running at http://localhost:3000");
```

👉 Save karo

🪜 STEP 4: Terminal open karo

```
Ctrl + `
```

🪜 STEP 5: Run karo (Node.js thi)

```
node server4.js
```

🪜 STEP 6: Browser ma test karo 🌐

👉 Ek-ek kari try kar:

🏠 Home Page

```
http://localhost:3000/
```

ℹ️ About Page

```
http://localhost:3000/about
```

👤 User Data

```
http://localhost:3000/user?name=raj&city=Ahmedabad
```

📁 Create File

```
http://localhost:3000/create
```

📖 Read File

```
http://localhost:3000/read
```

❌ Delete File

```
http://localhost:3000/delete
```

🎯 OUTPUT su avse?

👉 Different URL → different output
👉 File system pan kaam karse

🧠 Code samjuti (easy breakdown)

🔹 1. Modules

```
http, url, fs
```

👉 Server + URL + File system

🔹 2. Routing

```
if / else if
```

👉 URL pramane page change

🔹 3. Query

```
q.name, q.city
```

👉 URL mathi data levu

🔹 4. File operations

- create → `writeFile`
- read → `readFile`
- delete → `unlink`

❗ Important trick (exam mate)

👉 `/read` ma:

```
return;
```

✔ Etle double `res.end()` na thay

⚠️ Common mistake

❌ `/read` ma `return` bhuli jav
❌ file create pehla read karvu
❌ URL wrong lakhvu

🧠 Real life samjan

👉 Aa ek mini website + backend che

Feature	Use
/	Home
/about	Info
/user	Form data
/create	File create
/read	File read
/delete	File delete

🚀 Short summary

1. `server4.js` banav
2. Full code lakh
3. `node server4.js` run
4. Browser ma different URL try kar

🔥 Bhai aa practical perfect kari lidho to tu exam ma top level answer api sake 💯

Have bol — next JSON vala (25 thi start) pan sikhavu same style ma? 😎

Next

Maja aa gayi bhai 😎🔥 chalo have JSON vala practical start kariye (25 thi) — aa bahu easy ane scoring che 💯

🟢 PRACTICAL 25 – Display Employee Info using JSON

🧠 Aa practical ma su che?

👉 JSON object banavsu
👉 HTML ma display karisu

➡️ Aa Node.js nathi ❗
👉 Direct browser ma run thase

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: File banavo

```
emp.html
```

🪜 STEP 3: Code lakho

```html
<html>
<body>

<h2>Employee Details</h2>
<p id="demo"></p>

<script>
let emp = {
    "name": "Amit",
    "department": "IT",
    "salary": 45000
};

document.getElementById("demo").innerHTML =
emp.name + " works in " + emp.department + " department";
```

```
</script>

</body>
</html>
```

👉 Save karo

🪜 STEP 4: Run kem karvu?

👉 Simple 👇

- File par double click karo
  OR
- Right click → Open with Chrome

🎯 OUTPUT su avse?

👉 Screen par avse:

```
Employee Details

Amit works in IT department
```

🧠 Code samjuti (easy language ma)

🔹 1. JSON object

```
let emp = { "name":"Amit" }
```

👉 Data store karva mate

🔹 2. HTML ma show

```
document.getElementById("demo").innerHTML
```

👉 HTML element ma data mukva mate

🔹 3. Data use

```
emp.name
emp.department
```

👉 JSON mathi value levu

❗ Important note

👉 JSON format:

```
"key" : "value"
```

⚠️ Common mistake

❌ id match na hovu (`demo`)
❌ script bahar lakhvu
❌ quotes bhuli jav

🧠 Real life example

👉 JSON = data storage 📦
👉 HTML = display screen 📺

🚀 Short summary

1. `emp.html` banav
2. JSON object lakh

3. `innerHTML` thi show kar
4. Browser ma open

🔥 Bhai aa bahu easy che — exam ma direct lakhine marks pakka

Have bol — next Practical 26 (Student Marks JSON) sikhavu? 😎

Next

Chalo bhai 😎 next aavi gayu — Practical 26 (Student Marks using JSON)
👉 Aa 25 jevu j che, bas thodu data change 🔥

🟢 PRACTICAL 26 – Display Student Marks using JSON

🧠 Aa practical ma su che?

👉 JSON object ma student info
👉 HTML ma display karvu

➡️ Direct browser ma run (Node.js nathi)

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: File banavo

```
student.html
```

🪜 STEP 3: Code lakho

```html
<html>
<body>

<h2>Student Mark</h2>
<p id="result"></p>

<script>
let student = {
    "name": "vatsal",
    "subject": "web - 2",
    "marks": 90
};

document.getElementById("result").innerHTML =
student.name + " is scored " + student.marks;
</script>

</body>
</html>
```

👉 Save karo

🪜 STEP 4: Run kem karvu?

👉 Simple:

- Double click file

  OR
- Right click → Open with Chrome

🎯 OUTPUT su avse?

```
Student Mark

vatsal is scored 90
```

🧠 Code samjuti (easy language ma)

🔹 JSON object

```
let student = { name:"vatsal" }
```

👉 Student data store kare che

🔹 Data display

```
document.getElementById("result").innerHTML
```

👉 HTML ma data mukva mate

🔹 Access data

```
student.name
student.marks
```

❗ Important note

👉 JSON = JavaScript Object (data format)

👉 Keys always quotes ma hoy

⚠️ Common mistake

❌ id match na hovu (`result`)

❌ spelling mistake (`marks`)

❌ save na karvu

🧠 Real life samjan

👉 Student record system 🎓
👉 Data JSON ma store → screen par show

🚀 Short summary

1. `student.html` banav
2. JSON object lakh
3. `innerHTML` thi show kar
4. Browser ma open

🔥 Easy marks practical che aa 😎

Have bol — next Practical 27 (Product table JSON) sikhavu? (thodu advance ane mast lage che 😏)

Next

Chalo bhai 😎 have Practical 27 (Product Table using JSON) — aa thodu mast lage che 🔥
👉 Aa ma tame table banaavi ne data show karso

🟢 PRACTICAL 27 – Display Product Details in Table using JSON

🧠 Aa practical ma su che?

👉 Multiple products (array ma)
👉 Table ma display karvu

➡️ Thodu advance (loop use karisu 💪)

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: File banavo

```
product.html
```

🪜 STEP 3: Code lakho

```html
<html>
<body>

<h2>Product List</h2>

<table border="1">
<tr>
    <th>Name</th>
    <th>Price</th>
</tr>
<tbody id="table"></tbody>
</table>

<script>
let products = [
    {"name":"laptop","price":50000},
    {"name":"mobile","price":20000}
];

let rows = "";

products.forEach(p => {
    rows += "<tr><td>" + p.name + "</td><td>" + p.price + "</td></tr>";
});

document.getElementById("table").innerHTML = rows;
</script>
```

```
</body>
</html>
```

👉 Save karo

🪜 STEP 4: Run kem karvu?

👉 Simple:

- Double click file
  OR
- Right click → Open with Chrome

🎯 OUTPUT su avse?

👉 Table avse:

Name	Price
laptop	50000
mobile	20000

🧠 Code samjuti (easy language ma)

🔹 1. JSON Array

```
let products = [ {...}, {...} ]
```

👉 Multiple objects store kare che

🔹 2. Loop

```
products.forEach(p => {})
```

👉 Badha products par loop chale

🔹 3. Table rows banavvu

```
rows += "<tr>...</tr>"
```

👉 Dynamic HTML create kare

🔹 4. Display

```
innerHTML
```

👉 Table ma data muki de

❗ Important note

👉 `<tbody id="table">` bahu important che
👉 Same id use karvu pade

⚠️ Common mistake

❌ `<tbody>` ma id na mukvu
❌ loop na samajvu
❌ quotes mismatch

🧠 Real life example

👉 Amazon / Flipkart product list 🛒
👉 JSON data → table ma show

🚀 Short summary

1. `product.html` banav
2. JSON array lakh
3. loop use kar
4. table ma show

🔥 Bhai aa practical thi tu dynamic data display samji gayo 😎

Have bol — next Practical 28 (Button click JSON) sikhavu? (interactive che 🔘)

Mext

Chalo bhai 😎 next — Practical 28 (Button Click JSON)
👉 Aa thodu interactive che 🔘 (button click karta data show thase 🔥)

🟢 PRACTICAL 28 – Display JSON Data using Button Click

🧠 Aa practical ma su che?

👉 Button click karso
👉 Pachi JSON data screen par show thase

➡️ Event handling + JSON 💪

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: File banavo

```
button.html
```

🪜 STEP 3: Code lakho

```html
<html>
<body>

<button onclick="show()">Show Data</button>
<p id="data"></p>

<script>
function show(){
    let user = {
        "name":"Neha",
        "City":"Surat"
    };

    document.getElementById("data").innerHTML =
    user.name + " lives in " + user.City;
}
</script>

</body>
</html>
```

👉 Save karo

🪜 STEP 4: Run kem karvu?

👉 Simple:

- Double click file

  OR
- Right click → Open with Chrome

🎯 OUTPUT su avse?

👉 Page open thase
👉 Button dekhase: Show Data

👉 Button click karta 👇

```
Neha lives in Surat
```

🧠 Code samjuti (easy language ma)

🔹 1. Button

```
<button onclick="show()">
```

👉 Click karta function call thase

🔹 2. Function

```
function show()
```

👉 Button press thay tyare chale

🔹 3. JSON data

```
let user = { name:"Neha" }
```

👉 Data store kare che

🔹 4. Display

```
innerHTML
```

👉 Screen par data muki de

❗ Important note

👉 `onclick="show()"` must
👉 Function name same hovu joiye

⚠️ Common mistake

❌ function name mismatch
❌ id wrong (`data`)
❌ click na karvu 😄

🧠 Real life example

👉 Button = action 🔘
👉 Data = result 📊

Example:

- “Show Profile”
- “Get Details”

🚀 Short summary

1. `button.html` banav
2. button + function lakh
3. JSON data use kar
4. click → output

🔥 Bhai have tu interactive webpage banaavi sakhe che 😎

Have bol — next Practical 29 (Multiple Students Loop) sikhavu? (simple pan important che 👍)

Next

Chalo bhai 😎 next — Practical 29 (Multiple Students using JSON Loop)
👉 Aa simple che pan exam ma useful che 💯

🟢 PRACTICAL 29 – Display Multiple Students using JSON Loop

🧠 Aa practical ma su che?

👉 Multiple students (array ma)
👉 List (`<ul>`) ma display karvu

➡️ Loop concept strong banse 💪

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: File banavo

```
students.html
```

🪜 STEP 3: Code lakho

```html
<html>
<body>

<h2>Student List</h2>
<ul id="list"></ul>

<script>
let Student = [
    {"name":"Hevin"},
    {"name":"Henil"},
    {"name":"Vatsal"}
];

let output = "";

Student.forEach(s => {
    output += "<li>" + s.name + "</li>";
});

document.getElementById("list").innerHTML = output;
</script>

</body>
</html>
```

👉 Save karo

🪜 STEP 4: Run kem karvu?

👉 Simple:

- Double click file

  OR
- Right click → Open with Chrome

🎯 OUTPUT su avse?

👉 List avse:

- Hevin
- Henil

- Vatsal

🧠 Code samjuti (easy language ma)

🔹 1. JSON Array

```
let Student = [ {...}, {...} ]
```

👉 Multiple students store kare che

🔹 2. Loop

```
Student.forEach(s => {})
```

👉 Ek-ek student par chale

🔹 3. List banavvu

```
output += "<li>" + s.name + "</li>";
```

👉 `<li>` tag thi list item create kare

🔹 4. Display

```
innerHTML
```

👉 `<ul>` ma data muki de

❗ Important note

👉 `<ul id="list">` jaruri che
👉 Same id use karvu pade

⚠️ Common mistake

❌ id mismatch
❌ `<li>` tag bhuli jav
❌ loop wrong lakhvu

🧠 Real life example

👉 Student attendance list 📋
👉 JSON data → list ma show

🚀 Short summary

1. `students.html` banav
2. JSON array lakh
3. loop use kar
4. `<ul>` ma show

🔥 Bhai aa thi tu list rendering pan samji gayo 😎

Have last ma baki che:
👉 Practical 30 (Photo + JSON)
👉 pachhi 31 & 32 (Fetch + AJAX project) — thoda important 🔥

Bol next sikhavu? 😄

Next

Chalo bhai 😎 next — Practical 30 (JSON + Photo Display)
👉 Aa thodu stylish che 🔥 (image + data show karisu)

🟢 PRACTICAL 30 – Display Name & Photo using JSON

🧠 Aa practical ma su che?

👉 JSON ma:

- Name
- Photo (image file)

👉 HTML ma display karvu

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: File banavo

```
profile.html
```

🪜 STEP 3: Code lakho

```
<html>
<body>

<div id="profile"></div>
```

```
<script>
let person = {
    "name": "Family Photo",
    "photo": "photo.jpg"
};

document.getElementById("profile").innerHTML =
"<h3>" + person.name + "</h3>" +
"<img src='" + person.photo + "' width='200'>";
</script>

</body>
</html>
```

👉 Save karo

🪜 STEP 4: Important (Image file mukvi)

👉 Folder (`web2`) ma ek image mukjo

👉 Name exactly same hovu joiye:

```
photo.jpg
```

⚠️ Jo name alag hoy → image show nahi thay

🪜 STEP 5: Run kem karvu?

👉 Simple:

- Double click file

  OR
- Right click → Open with Chrome

🎯 OUTPUT su avse?

👉 Screen par avse:

- Family Photo (heading)
- niche image show thase 📸

🧠 Code samjuti (easy language ma)

🔹 1. JSON object

```
let person = { name:"Family Photo" }
```

👉 Data store kare che

🔹 2. Image display

```
<img src='photo.jpg'>
```

👉 Image show kare che

🔹 3. Dynamic HTML

```
innerHTML
```

👉 Name + image combine kari ne show kare

❗ Important note

👉 Image file same folder ma hovi joiye
👉 File name exact match

⚠️ Common mistake

❌ image path wrong
❌ spelling mistake (`photo.jpg`)
❌ file folder ma na hoy

🧠 Real life example

👉 Profile card 👤
👉 Name + photo show

🚀 Short summary

1. `profile.html` banav
2. JSON ma name + photo path
3. image folder ma muk
4. browser ma open

🔥 Bhai have JSON na badha basics complete 😎

Have last 2 important baki che:
👉 Practical 31 (Fetch JSON file)
👉 Practical 32 (AJAX Mini Project 💥)

Aa thoda exam + viva ma important che 🔥

Bol next (31) sikhavu? 😄

Next

Chalo bhai 😎 now Practical 31 (Fetch JSON from external file) — aa thodu important che 🔥 (real world ma bahu use thay che)

🟢 PRACTICAL 31 – Display JSON from External File using fetch()

🧠 Aa practical ma su che?

👉 JSON alag file ma hase
👉 HTML thi fetch kari ne data show karso

➡️ Real API concept ni starting 💪

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: 2 file banavo

📄 File 1:

```
index.html
```

📄 File 2:

```
file2.json
```

🪜 STEP 3: JSON file ma code lakho

👉 `file2.json`

```
{
  "name": "Vatsal"
}
```

👉 Save karo

🪜 STEP 4: HTML file ma code lakho

👉 `index.html`

```html
<html>
<body>

<h2>External JSON</h2>
<p id="demo"></p>

<script>
fetch("file2.json")
.then(res => res.json())
.then(data => {
    document.getElementById("demo").innerHTML = data.name;
});
</script>

</body>
</html>
```

👉 Save karo

🪜 STEP 5: Run kem karvu? ⚠️ IMPORTANT

👉 Aa direct double click thi nahi chale ❌

👉 Use karvu pade:

👉 VS Code extension: Live Server

Steps:

1. Right click `index.html`
2. Click → Open with Live Server

🎯 OUTPUT su avse?

👉 Screen par avse:

```
External JSON

Vatsal
```

🧠 Code samjuti (easy language ma)

🔹 1.

```
fetch("file2.json")
```

👉 JSON file ne call kare che

🔹 2.

```
res.json()
```

👉 Data ne JSON format ma convert kare

🔹 3.

```
data.name
```

👉 JSON mathi value levu

🔹 4.

```
innerHTML
```

👉 Screen par show kare

❗ Important note

👉 Direct open nahi thay (CORS issue)
👉 Live Server jaruri

⚠️ Common mistake

❌ file name wrong (`file2.json`)
❌ Live Server use na karvu
❌ path wrong

🧠 Real life example

👉 Website API call kare che 🌐
👉 Data fetch kari ne display kare

🚀 Short summary

1. `file2.json` banav
2. `index.html` ma fetch use kar
3. Live Server thi run kar
4. data show

🔥 Bhai aa samji gayo to tu API fetch concept pakdi lidho 💪

Have last ane sabthi important 🔥
👉 Practical 32 (AJAX Mini Project) — aa exam ma mast impact padse 💥

Bol sikhu? 😎

Practical 32

Chalo bhai 😎🔥 aa last ane sabthi important — Practical 32 (AJAX Mini Project)
👉 Aa full mini project che 💥 (exam + viva mate top level)

🟢 PRACTICAL 32 – AJAX Student Info Project

🧠 Aa project ma su che?

👉 User ID enter kare
👉 Button click kare
👉 JSON file mathi data fetch thai ne show thay

➡️ Real-time data system jevu 💪

🪜 STEP 1: Visual Studio Code open karo

👉 Same folder (`web2`)

🪜 STEP 2: 4 file banavani che

```
index.html
script.js
students.json
style.css
```

🪜 STEP 3: index.html code

```html
<html>
<head>
<title>AJAX Student Info</title>
<link rel="stylesheet" href="style.css">
</head>

<body>
```

```html
<h2>Student Information System (AJAX)</h2>

<input type="text" id="sid" placeholder="Enter Student ID">
<button onclick="getStudent()">Get Details</button>

<div id="result"></div>

<script src="script.js"></script>

</body>
</html>
```

🪜 STEP 4: script.js code

```javascript
function getStudent() {

    var sid = document.getElementById("sid").value;

    var xhr = new XMLHttpRequest();

    xhr.open("GET", "students.json", true);

    xhr.onreadystatechange = function () {

        if (xhr.readyState == 4 && xhr.status == 200) {

            var data = JSON.parse(xhr.responseText);

            var students = data.students;

            var output = "Student Not Found";

            for (var i = 0; i < students.length; i++) {

                if (students[i].id == sid) {

                    output = `
                    Name: ${students[i].name} <br>
                    Course: ${students[i].course} <br>
                    Marks: ${students[i].marks}
                    `;
                }
            }
```

```
            document.getElementById("result").innerHTML = output;
        }
    };

    xhr.send();
}
```

🪜 STEP 5: students.json code

```
{
  "students": [
    {"id": "101", "name": "Vatsal", "course": "CS", "marks": 85},
    {"id": "102", "name": "Hevin", "course": "IT", "marks": 90},
    {"id": "103", "name": "Henil", "course": "AI", "marks": 88}
  ]
}
```

🪜 STEP 6: style.css (optional design)

```
body {
    font-family: Arial;
    text-align: center;
    background: #f4f4f4;
}

input, button {
    padding: 10px;
    margin: 10px;
}

#result {
    margin-top: 20px;
    font-size: 18px;
}
```

🪜 STEP 7: Run kem karvu? ⚠️ IMPORTANT

👉 Direct double click nahi ❌

👉 Use:

👉 Live Server

Steps:

1. Right click → `index.html`
2. Click → Open with Live Server

🎯 OUTPUT su avse?

👉 Screen par:

- Input box
- Button

👉 Example:

- ID enter karo: `101`
- Click "Get Details"

👉 Output:

```
Name: Vatsal
Course: CS
Marks: 85
```

🧠 Code samjuti (simple language ma)

🔹 1. Input levu

```
document.getElementById("sid").value
```

🔹 2. AJAX request

```
XMLHttpRequest()
```

👉 Server thi data fetch kare

🔹 3. JSON parse

```
JSON.parse()
```

👉 String → object convert

🔹 4. Loop

```
for loop
```

👉 Student search kare

🔹 5. Output

```
innerHTML
```

❗ Important points (exam mate 🔥)

✔ AJAX = asynchronous data loading

✔ `readyState == 4` → complete

✔ `status == 200` → success

✔ JSON file thi data fetch

⚠️ Common mistake

❌ Live Server use na karvu
❌ file name mismatch
❌ id wrong
❌ JSON format wrong

🧠 Real life example

👉 College portal 🎓
👉 ID enter → student detail show

🚀 Final summary (Full syllabus complete 🎉)

👉 Node.js basics ✔️
👉 Server ✔️
👉 Routing ✔️
👉 File System ✔️
👉 JSON ✔️
👉 Fetch ✔️
👉 AJAX ✔️

🔥 Bhai tu aapde badha practical cover kari lidha 💯

👉 Have jo tu kahe to hu:

- Viva questions & answers api dau
- Athva badha practical nu short revision notes banaavi dau

Bol su joiye 😎

Messages beyond this point are only visible to you

How can I download this chat as pdf

Good question 👍 You have 3 easy ways to download this chat as PDF.

✅ METHOD 1 (Easiest – Browser Print)

Works in Google Chrome, Microsoft Edge, etc.

Steps:

1. Scroll up and load full chat
2. Press:

```
Ctrl + P
```

3. In print options:
   - Destination → Save as PDF
4. Click Save

👉 Your chat will download as PDF

✅ METHOD 2 (Mobile Phone 📱)

If you are using phone:

Steps:

1. Tap 3 dots menu (browser)
2. Click Share / Print
3. Select:
   👉 Save as PDF

✅ METHOD 3 (Copy-Paste to Word)

Use Microsoft Word or Google Docs

Steps:

1. Copy full chat
2. Paste in Word / Docs
3. Click:
    - File → Save as PDF

🔥 PRO TIP (Best Result)

👉 Before saving:

- Zoom out (Ctrl -) → more content in one page
- Remove unwanted messages if needed

💡 BONUS (If you want)

I can also:
👉 Convert all practicals into clean PDF notes
👉 Proper headings + formatting (exam ready 📘)

Just tell me:
"Make PDF notes" 😎